Hence the minimum p.d to excite electrons must be [illegible] V.

27. Ans: D

Heisenberg's uncertainty principle, $\Delta p \Delta x \geq h$

Minimum uncertainty in the momentum, $\Delta p = m\Delta v = \dfrac{h}{\Delta x}$

Minimum uncertainty in the speed, $\Delta v = \dfrac{h}{m\Delta x}$

k.e. = $\frac{1}{2} mv^2$

$$\text{Minimum } \frac{\Delta \text{k.e.}}{\text{k.e.}} = 2 \text{ x minimum } \frac{\Delta v}{v}$$

$$= \frac{2h}{mv\Delta x}$$

$$= \frac{2h}{p\Delta x}$$

$$= \frac{2 \times 6.63 \times 10^{-34}}{2 \times 10^{-22} \times 1 \times 10^{-10}}$$

$= 0.07$ or 7%

28 Ans: A